Is a Production Volatility-Targeting Rule an Alpha? A Sealed, Pre-Registered Test on a Century of Factor Returns

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Abstract

Volatility-managed portfolios earn large spanning alphas (Moreira and Muir, 2017). Those alphas are hard to capture in real time (Cederburg et al., 2020), and part of them reflects a leverage constant estimated with look-ahead (Liu et al., 2019). We ask a narrower question with a stricter design: does a specific, already-deployed volatility-targeting overlay earn alpha out of the era it was designed in? The overlay combines a HAR volatility-forecast lever capped at 1.5×, CVaR and drawdown brakes, and a momentum tilt, and it is causal by construction. We pre-registered the test, data, tolerances and decision rule; first replicated the published results within stated tolerances; then sealed the 1926–2012 holdout by committing a hash of the code before running it once.

On the market, the overlay raises the Sharpe ratio from 0.39 to 0.53 (Ledoit–Wolf one-sided $p = 0.047$), and its univariate spanning alpha is 2.67% a year ($t = 2.94$). Net of the Fama–French five factors, momentum and betting-against-beta, the alpha is −1.19% ($t = -1.18$, 1963-07–2012-12). The managed market loads on profitability, momentum and BAB, and the pre-registered verdict is that no alpha survives.

Applied to factor portfolios, the same real-time rule earns gross full-model alphas that survive Romano–Wolf correction for HML, RMW, CMA, momentum and BAB. These are gross of the factors' own trading costs, which factor returns cannot price. We state what stock-level CRSP and TAQ would change and stop there.

JEL: G11, G12, C12, C58. **Keywords:** volatility targeting, volatility-managed portfolios, pre-registration, spanning tests, multiple testing, backtest overfitting.

1 Introduction

A volatility-targeting rule scales exposure to a risky portfolio inversely to a forecast of its volatility. Fleming et al. (2001) document the economic value of volatility timing in asset allocation. Barroso and Santa-Clara (2015) show that scaling momentum by its own realized variance roughly doubles its Sharpe ratio and removes its crashes (see also Daniel and Moskowitz, 2016). Harvey et al. (2018) find that volatility targeting raises Sharpe ratios for equities and reduces the probability of extreme returns.

Moreira and Muir (2017) make the strongest form of the claim. They scale a factor by $c/\hat{\sigma}^2_{t-1}$, where $\hat{\sigma}^2_{t-1}$ is last month's realized variance, and regress the result on the unmanaged factor. The spanning alpha is large for the market and for most factors.

The literature since has pushed back on whether investors can collect it. Cederburg et al. (2020) build the strategies an investor could have run in real time. They find that combining the managed and original factor improves on the original for some factors, momentum and BAB

above all, but not for the market. Liu et al. (2019) point out that the constant c is chosen on the full sample. With c estimated in real time, they report maximum drawdowns that are deeper than the market's.

We study one deployed rule rather than the textbook one. It is the risk overlay of an open-source quantitative platform, and it was designed and tuned on 2013–2026 data. It differs from the Moreira and Muir (2017) rule in ways a practitioner would recognize:

- it targets an annualized volatility of 13% using a causal HAR forecast (Corsi, 2009), refit monthly;
- its leverage is capped at 1.5×;
- CVaR and drawdown brakes cut exposure after losses;
- the result is smoothed over five days;
- a ±15% tilt follows the sign of the trailing 21-day return.

Every input is trailing, so the rule needs no full-sample constant.

The question is whether this rule earns alpha, net of the factors a referee would name, on years it has never seen.

Three design choices make the answer credible.

1. *Pre-registration and a seal.* Before any file was downloaded, we committed:
   - the question and the data;
   - replication targets with numeric tolerances;
   - the test battery;
   - a four-part decision rule.

   The 1926–2012 holdout was then sealed. The SHA-256 of the exact code that would produce the verdict was committed before that code first touched a pre-2013 return, and the code was run once.

2. *Replicate first.* The program was allowed to proceed only if the published results of Moreira and Muir (2017) and Cederburg et al. (2020) reproduced within the registered tolerances on today's data. They did.

3. *Survivorship-free test assets.* The overlay is a portfolio-level timing rule, so we test it on portfolios rather than on a hand-picked stock universe. We use Ken French's market and factor portfolios, built from CRSP from 1926, and AQR's BAB factor.

Findings.

- *The market.* On the market, the overlay is a real Sharpe-ratio improvement for an investor who holds only the market: 0.53 against 0.39 over 1927-07–2012-12. It is not an alpha. Net of FF5, momentum and BAB the alpha is -1.19% a year ($t = -1.18$). The managed market's extra return is explained by its loadings on profitability (RMW), momentum and BAB. Most of the univariate alpha accrues before 1963.

- *Costs.* Trading costs do not decide this result. The breakeven cost on the univariate alpha is 51 bp per unit of turnover, against about 0.21 bp for half an E-mini tick today.
- *Factor portfolios.* On factor portfolios the same rule earns large gross alphas in real time, and they survive Romano–Wolf correction. We do not claim them net of costs.

2 Data

Sources.

- *Factors.* Daily and monthly Fama–French research factors and momentum from Ken French's data library, built from the 202608 CRSP database: Mkt−RF, SMB, HML and RF from 1926-07; RMW and CMA from 1963-07 (Fama and French, 2015); UMD from 1926-11 (Carhart, 1997).
- *BAB.* AQR's daily US betting-against-beta factor, from 1930-12 (Frazzini and Pedersen, 2014).
- *VIX.* Cboe's VIX, used only descriptively.

Archiving. French revises history, so a URL does not identify a dataset. Every file was fetched once, after the registration was committed, and is pinned by SHA-256 in a committed manifest. Every loader verifies the hash before parsing and refuses a mismatch. The raw files are not redistributed. Anyone reproducing either obtains the same bytes or is told that they hold a different vintage.

CSV floats are parsed with correctly rounded (round-trip) conversion. The default parser in our stack rounds through long double, which differs between x86 and arm64 hardware.

Financing. The factors are excess returns over French's one-month T-bill rate RF. Holding exposure e in Mkt−RF is the same as holding e in the market and $1 - e$ in bills, or borrowing $e - 1$ at the bill rate. Leverage is therefore financed at the T-bill rate. A robustness check finances it at the three-month bill (DTB3) instead.

Monthly returns. The overlay rebalances daily. The monthly excess return of a position with daily exposure e_t is

$$R = \prod_t (1 + e_{t-1} f_t + r_t^f) - \prod_t (1 + r_t^f),$$

taken over a calendar month's trading days. Every managed and unmanaged series in the Stage-2 tests is built this way, so no comparison mixes French's monthly files with daily-compounded returns.

Saturdays and holidays. The NYSE traded on Saturdays until 1952, and the overlay's windows assume weekday sessions. Each Saturday return is therefore compounded into the next observation. AQR's calendar lists 382 US holidays from 1984 on that French's calendar omits; BAB returns on those days are compounded forward the same way. The replication uses French's files unchanged, as the replicated papers did.

3 Design

3.1 Windows and the seal

- *Holdout:* 1926-07 to 2012-12.
- *Full-model span:* 1963-07 to 2012-12, the span over which all seven regressors exist.
- *Contaminated window:* 2013-01 onward. The overlay was designed on these years, so they are reported descriptively and enter no decision.

The registration, an amendment fixing the replication samples, and the seal were committed in that order, each before the computation it governs. The seal amendment records the code digest. A verification gate refuses any results file whose recorded digest differs.

3.2 Stage 1: the replication gate

Construction. The managed factor is $f_t^{\sigma} = (c/\mathrm{RV}_{t-1}^2) f_t$, where:

- RV^2 is the demeaned sum of squared daily returns within the month;
- c equates the standard deviations of f^{σ} and f over the sample.

Targets and tolerances. We register the Table 1 spanning alphas of Moreira and Muir (2017), with heteroskedasticity-robust standard errors. The samples match their reported N exactly: they end in 2015-04 and start in 1926-08, 1927-01 or 1963-08.

The tolerance on $|\Delta\alpha|$ is 0.75 SE for the market and momentum, plus $|\Delta\beta| \leq 0.06$ for those two; it is 1 SE for SMB, HML, RMW and CMA. The tolerances allow for French's revisions since the paper was written.

From Cederburg et al. (2020) we register:

- the in-sample Sharpe ratios (their Table 1);
- the real-time combination strategy (their Table 5), built as follows:
  - an expanding window that starts after $K = 120$ months;
  - c_t estimated on the training window;
  - mean–variance weights at $\gamma = 5$;
  - the position clipped at $|y_t| \leq 5$.

Pass rule. The gate passes if:

- the market and momentum both pass;
- at least three of the other four factors pass;
- all three Cederburg et al. market checks pass.

If the gate failed, the program stopped.

The look-ahead point. The full text of Liu et al. (2019) was not available to us. We register only their abstract's claim, as a descriptive check: maximum drawdowns of roughly 68–93% once c is real-time. We reproduce the construction directly.

3.3 Stage 2: the overlay

Primary. The primary strategy is the overlay on the market, exactly as deployed:

- *Lever:* target 13%, HAR forecast, cap 1.5×.
- *CVaR brake:* target 1.8% daily at 95%.
- *Drawdown brake:* a linear ramp from −8% to −20%, with a floor of 0.25.
- *Smoothing:* 5 days.
- *Tilt:* ±15%.

Its estimators read the market's total return, as they read the deployed book's. Exposure decided at the close of day t applies to day $t + 1$. The first 252 observations train the estimators and are not evaluated.

A verification gate shows that this code reproduces the deployed backtest's applied exposure exactly, on every live day of a synthetic panel.

One property of the rule was known in advance and registered. The drawdown brake reads the market's own high-water mark, so after 1929 it holds near its floor until that mark is regained.

Family. The family applies the same rule to SMB, HML, RMW, CMA, UMD and BAB. The deployed thresholds are denominated in the deployed book's volatility, so each factor is first expressed in those units: $\tilde{f}_t = f_t \cdot 0.13/\hat{\sigma}_{t-1}$. Here $\hat{\sigma}$ is the factor's trailing realized volatility over 2,520 observations, expanding from a 252-observation minimum. No alternative normalization was tried.

Tests. All regressions are monthly, with Newey and West (1987) standard errors at 12 lags and alphas annualized. The battery is:

- univariate spanning regressions on each managed factor's own unmanaged version;
- full-model regressions on Mkt, SMB, HML, RMW, CMA, UMD and BAB;
- the Ledoit and Wolf (2008) studentized circular-block bootstrap test of the Sharpe-ratio difference (12-month blocks, 5,000 resamples, one-sided);
- the Romano and Wolf (2005) studentized stepdown across the seven full-model alphas, at 5% family-wise;
- the SPA test of Hansen (2005) on the factor-hedged series, using the stationary bootstrap of Politis and Romano (1994);
- univariate alphas in four eras;
- an Andrews (1993) sup-Wald test for a break in the intercept, with its p-value simulated from Brownian bridges.

Table 1: Moreira–Muir Table 1 on the 202608 French vintage. Spanning regression $f^{\sigma} = \alpha + \beta f + \varepsilon$, monthly, α in % a year, White SEs in parentheses. Tolerance: the registered bound on $|\Delta\alpha|$.

	Published			Replication				
Factor	α	(SE)	β	α	(SE)	β	N	Tol.
Mkt	4.86	(1.56)	0.61	4.93	(1.57)	0.60	1,065	1.17
UMD	12.51	(1.71)	0.47	12.83	(1.72)	0.45	1,060	1.28
SMB	−0.58	(0.91)	0.62	−0.59	(0.91)	0.60	1,065	0.91
HML	1.97	(1.02)	0.57	1.73	(1.05)	0.54	1,065	1.02
RMW	2.44	(0.83)	0.62	2.80	(0.89)	0.58	621	0.83
CMA	0.38	(0.67)	0.68	0.36	(0.67)	0.67	621	0.67

Overfitting diagnostics. A registered eight-member grid of deployed-rule siblings (tilt, cap and brakes, each on or off) supplies:

- a CSCV probability of backtest overfitting (Bailey et al., 2017);
- the dispersion for a deflated Sharpe ratio (Bailey and López de Prado, 2014).

Every member and every family factor is charged to the platform's cumulative trial ledger, which now stands at $N = 202$.

Decision rule. The overlay earns alpha if and only if, for the market, all four of the following hold:

1. the full-model alpha is positive with $t \geq 3$, the hurdle of Harvey et al. (2016);
2. the Ledoit–Wolf $p < 0.05$;
3. the alpha survives Romano–Wolf;
4. the univariate alpha is positive in at least three of four eras.

4 Stage 1: the published results replicate

Table 1 compares the replication with Moreira and Muir (2017). Every factor lands inside its tolerance, most by a wide margin. The largest gap relative to its tolerance is RMW's: 2.80 against a published 2.44.

The Cederburg et al. (2020) checks also pass.

- *In sample.* The managed market's Sharpe ratio is 0.52 against 0.42, as published (0.51 against 0.42). The momentum gain is +0.51 (published +0.50).
- *Real time, market.* The combination strategy over 1936-08–2016-12 has a Sharpe ratio of 0.44, against 0.46 for the original factor alone. The difference is −0.02, with $p = 0.80$ in the test of Jobson and Korkie (1981). The published values are 0.42, 0.46 and −0.04.
- *Real time, momentum.* The gain survives in real time: 0.93 against 0.48, as published (0.92 against 0.49).

Table 2: Univariate alpha of the managed market by era (% a year, Newey–West t).

Era	α	(t)	Months
1927–1945	4.31	(1.97)	222
1946–1962	1.59	(1.85)	204
1963–1987	0.76	(0.64)	300
1988–2012	1.04	(0.79)	300

The look-ahead point also reproduces. With c estimated in real time (1936-08–2016-12), the managed market's spanning alpha falls to 3.40% (SE 2.30, $t = 1.5$). Leverage reaches 18.9×, and the maximum drawdown is −90.8%, against −54.4% with the full-sample constant and −50.3% for the market. That lies inside the 68–93% range Liu et al. (2019) report.

The market alpha of a volatility-managed portfolio thus depends on the full-sample constant. The rule we test has no such constant.

5 Stage 2: the deployed overlay

5.1 The market

The verdict is **NO ALPHA SURVIVES**. Of the four registered criteria:

- the full-model $t \geq 3$ test fails;
- the Ledoit–Wolf test holds;
- Romano–Wolf survival fails;
- positive alpha in at least three of four eras holds.

For an investor holding only the market, the overlay is an improvement. Over 1927-07–2012-12 ($N = 1,026$ months) its Sharpe ratio is 0.53 against 0.39, a gain of +0.14 with Ledoit–Wolf one-sided $p = 0.047$. The univariate spanning alpha is 2.67% a year ($t = 2.94$, $\beta = 0.39$).

The gain is not an alpha with respect to the factors. Over 1963-07–2012-12, regressing the managed market on all seven factors gives $\alpha = -1.19\%$ ($t = -1.18$, $R^2 = 0.73$). The loadings are 0.62 on the market, 0.10 on RMW, 0.10 on momentum, 0.06 on BAB, 0.06 on CMA, 0.01 on SMB and −0.07 on HML. The Romano–Wolf adjusted p-value for the market is 0.86.

The loadings have a natural reading. The drawdown brake and the 21-day tilt cut exposure after the market falls, which is a time-series trend component and matches the momentum loading. De-levering when volatility is high is a defensive tilt, and it co-moves with profitability and BAB. Once an investor holds those factors, the overlay adds nothing.

Table 2 shows where the univariate alpha comes from. It is positive in all 4 eras and concentrated before 1963: only 1927–1945 reaches $t \approx 2$. In the two eras since 1963 it is about 1% a year with $t < 1$. The sup-Wald statistic is 8.67, at 1998-04, with $p = 0.051$.

The contaminated window, 2013-01–2026-08, is not part of the decision. It points the same way: the univariate alpha is −0.26% ($t = -0.2$), and the managed market trails the market (0.73 against 0.91).

Grid, PBO and DSR. Across the registered eight-member grid, holdout Sharpe ratios range from 0.457 to 0.530. The deployed member is the highest, at 0.530. The CSCV probability of backtest overfitting across the grid is 0.80, over 12,870 splits: the in-sample winner usually lands below the out-of-sample median. The choice among the grid's tilt, cap and brake settings therefore does not persist, and the deployed member's rank is not evidence for those choices.

The deflated Sharpe ratio at $N = 202$ is 1.00, with $\mathrm{SR}_0 = 0.106$ taken from the grid's dispersion. It is uninformative here: it tests whether the managed market's Sharpe ratio exceeds zero, and the market's own Sharpe ratio does.

For the record, the platform's repository-wide PBO is 0.45 over 140 variants.

Robustness.

- *Bill-rate financing.* Financing leverage at DTB3 instead of RF (from 1954, mean spread 0.35% a year) leaves the full-model alpha at -1.23% ($t = -1.22$).
- *The volatility forecast.* The overlay's HAR forecast tracks VIX closely: the daily correlation is 0.86 in 1990–2012 and 0.86 after.

5.2 Costs

The overlay turns over 5.25 units of exposure a year. Its mean exposure is 0.71, and it sits at the cap 0.8% of the time.

At the univariate alpha, the breakeven one-way cost is 51 bp per unit of turnover. Against the full-model alpha the breakeven is negative: no cost is low enough.

For comparison, the E-mini S&P 500 has a minimum price fluctuation of 0.25 index points on a \$50-per-point contract (CME Rulebook, chapter 358). Half a tick is $0.125/L$ of notional, about 0.21 bp at an index level L near 6,000.

Costs are therefore not what is missing on the market; the alpha is. Before 1982 no index future existed, and we do not model historical trading costs.

5.3 Factor portfolios

Table 3 reports the family. These results are reported under the registered Romano–Wolf correction and are not part of the decision.

- *Survivors.* The full-model alphas of HML, RMW, CMA, momentum and BAB survive at 5% family-wise. SMB's does not.
- *SPA.* Hansen's SPA test rejects the hypothesis that no managed factor has a positive alpha ($p = 0.0002$).
- *Sharpe ratios.* They rise substantially for momentum and BAB, and for HML and RMW.

The rules are real-time: the scale is causal and there is no full-sample constant. That distinguishes these results from the in-sample spanning alphas. The pattern, with momentum and BAB strongest, matches the real-time evidence of Cederburg et al. (2020) and the momentum results of Barroso and Santa-Clara (2015).

Two caveats bound what Table 3 supports.

Table 3: The deployed overlay on factor portfolios, each first expressed in the deployed volatility units. Alphas in % a year, Newey–West t. RW: Romano–Wolf stepdown-adjusted p over the seven full-model alphas (1965-07–2012-12, $N = 570$). SR: Sharpe ratio of the raw and managed factor over each factor's own span; LW p one-sided. Scale: median $0.13/\hat{\sigma}$ before the overlay's own lever.

	Univariate		Full model			Sharpe ratio			
Factor	α	(t)	α	(t)	RW p	Raw	Managed	LW p	Scale
SMB	0.68	(1.03)	0.99	(1.14)	0.209	0.15	0.21	0.199	1.6
HML	4.06	(3.68)	5.78	(4.00)	<0.001	0.43	0.61	0.014	2.0
RMW	5.50	(3.53)	7.43	(4.66)	<0.001	0.48	0.75	0.015	3.3
CMA	2.91	(2.82)	4.80	(3.54)	<0.001	0.60	0.71	0.083	2.7
UMD	7.88	(5.85)	8.81	(3.97)	<0.001	0.46	0.88	<0.001	1.6
BAB	8.57	(5.33)	12.18	(5.53)	<0.001	0.81	1.19	<0.001	1.9

1. *Levered units.* The alphas are in levered units: the median scale runs from 1.6× to 3.3× before the overlay's own lever. The t-statistics and Sharpe ratios are scale-free; the alpha magnitudes are not.

2. *No trading costs.* A long–short factor rebalances its stock legs every month and pays to short, and the overlay adds rescaling turnover on top. None of this can be priced from factor returns (compare Novy-Marx and Velikov, 2016; Barroso and Detzel, 2021).

We therefore report these as gross, real-time spanning alphas, and make no net-of-cost claim.

6 What the missing data would change

The study used only data an undergraduate can obtain without a WRDS subscription. Three missing datasets would sharpen specific results, and in each case we stop rather than substitute.

- *Stock-level CRSP* would let the overlay run on a point-in-time version of the deployed equal-weight book itself, rather than on the market as a proxy for it. It would also let the factor legs be rebuilt with their turnover, so that the family's alphas could be costed.

- *TAQ* would supply consolidated effective spreads against the NBBO, for SPY and for the factor legs. For the market overlay this cannot change the verdict, because no cost rescues a negative full-model alpha. For the family it is the missing half of a net-of-cost claim.

- *IEX HIST* trade messages would give a one-venue SPY spread. Measuring it requires accepting IEX's data terms, which we have not yet done. As above, it could not change the market verdict.

Two external checks are pending:

- the correlation of the overlay with S&P's published daily risk-control index levels;

- a broker paper account trading the overlay on SPY, which gives every order a third-party timestamp.

7 Conclusion

A capped, braked, forecast-driven volatility-targeting overlay, run unchanged on a century of survivorship-free data it was never tuned on, raises the market's Sharpe ratio. It does not earn alpha with respect to the factors an investor can already hold. On the market it is risk control.

On factor portfolios the same real-time rule produces gross alphas that survive multiple-testing correction. Whether they survive trading costs is a question for stock-level data.

A Reproducibility

The registration, both amendments, the code, the results files and the verification gates are in the public repository. The pre-registration is `docs/design/DESIGN29.md`, and its amendments record the sample correction and the seal.

The data manifest `data/vendor/MANIFEST.json` lists each vendor file's URL, retrieval time, size and SHA-256.

Running `run_research.py volalpha_replication` and `run_research.py volalpha` rebuilds both results files from the archived bytes. Two gates check the work:

- *X46* checks the replication machinery and applies the registered tolerances.
- *X47* checks the overlay identity and the size and power of every test. It also refuses a results file not produced by the sealed code.

Every number in this paper is a macro generated from those files.

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